

Motor Datasheet: WxF-50x30GT 1214 mk1

Machine Configuration

Parameter	Value	Unit
Machine Outer Diameter	48.0	mm
Machine Length / Thickness	29.5	mm
Voltage	48.0	V
Operating Temperature	-40 to 85	°C

Mass Properties

Parameter	Value	Unit
Machine Mass (active components only)	254.9	g
Rotor Mass	77.4	g
Stator Mass	177.5	g
Rotor Structure Included	No	-
Rotor Moment of Inertia	249.3	g·cm ²

Performance Metrics

Parameter	Value	Unit
Torque (continuous)	936.5	mNm
Speed	2500	rpm
Losses (continuous)	29.6	W
Supply Current (RMS)	4.99	A
Phase Resistance (per phase)	357.3	mΩ
Terminal Inductance ($L_{tt} = L_d + L_q$)	2157.0	μH
D-axis Inductance (L_d)	1103.1	μH
K_e (peak, phase)	8.48	mV/rpm
K_e (peak, line-to-line)	14.70	mV/rpm
K_t (peak)	122.3	mNm/A
Motor Constant (K_m)	181.4	mNm/√W